been trained against them. This thesis therefore treats representation and transfer as the primary questions, and treats online adaptation as the last of them rather than the first. The research statement of the thesis is as follows:

Comparison notation.

For readability, the three policy–actuator stacks are denoted by short labels throughout this thesis: *R* is PPO–ROM, *IF* is PPO–ID–FD, and *AIF* is PPO–(AAM)–ID–FD. R uses the command-conditioned reduced-order model; IF uses the frozen inverse/forward pair and AIF augments IF with online adaptation. The labels name the complete stacks, not just their actuator models.

> *To determine how hydraulic actuator dynamics learned from high-fidelity simulation can be reduced, transferred into a computationally efficient robotic simulation and used for reinforcement-learning control, and under what conditions online adaptation improves on the robustness those learned models already provide.*

The statement decomposes into six research questions, which are answered in the order given. The first three concern how the actuator is represented, the fourth concerns control through that representation, and the last two concern what remains for adaptation once the first four are settled. The six questions are:

1. How can data from a high-fidelity Simscape model be used to learn computationally tractable representations of hydraulic actuation while preserving the command dependence a controller requires?
2. To what extent do actuator models identified from Simscape transfer to an IsaacLab manipulator without deployment-specific retraining?
3. Can a physics-structured reduced-order hydraulic model supply dynamics that are accurate and stable enough for GPU-parallel policy training?
4. Can a policy trained from a privileged classical teacher become the sole deployed controller while acting through learned actuator dynamics?
5. How does the choice of actuator representation affect tracking, command behaviour, latency and robustness under plant variation?
6. Where plant variation survives that robustness, is it identifiable from recent observations, and does conditioning the controller on it improve closed-loop performance?

The first is answered in Chapters 4 and 5, the second and third in Chapter 5, the fourth in Chapter 6, and the last two in Chapter 7.

RQ6 is the question named in the title, and it is deliberately placed last. It is only worth asking once the preceding questions have established that the learned models transfer